Dwight Look College of
ENGINEERING
TEXAS A&M UNIVERSITY

Sound Localization Robot, Team 5

Bi-Weekly Update 4

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TA: Muhammad Bin Mughees
Sponsor: Dr. Moreira

Project Scope

Figure 1. First responder crawls through smoke and broken glass to rescue elderly man from house fire (Independent TV)

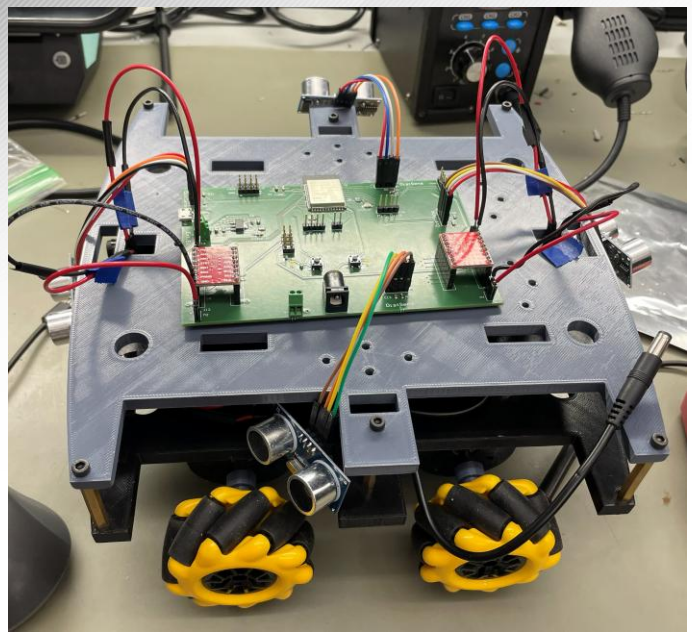
- Keeps **first responders safe** by entering hazardous or toxic areas first.
- **Reduces search and rescue time** by avoiding exploring unnecessary paths
- **Reliable in low-visibility** or high-stress conditions (smoke, debris, noise).

Problem: First Responders lack reliable methods for navigating dangerous environments, where visibility is limited, putting both their safety and the victims at risk.

Our Solution: Building a Sound Localization robot that can hear distress calls, pinpoint their direction, and move toward them autonomously, allowing first responders to focus on saving lives instead of risking theirs.

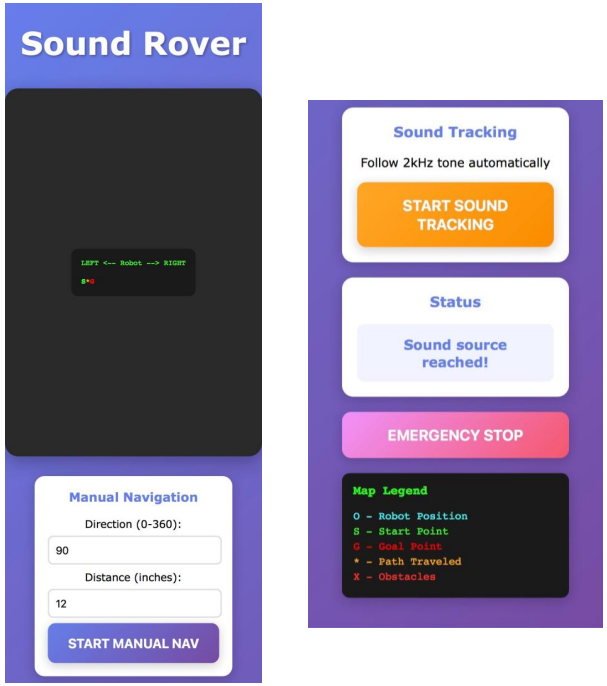
*A smarter, safer, and faster way to locate victims, all while protecting rescuers and saving more lives.

Integrated System Diagram:



Website generated through internal Wi-Fi

User Commands: Emergency Stop, Origin

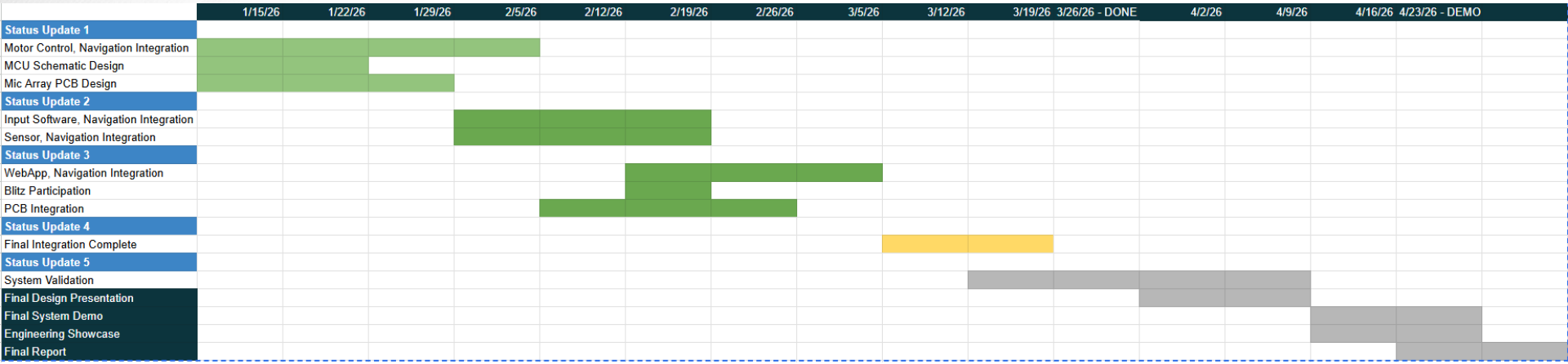


User Interface – viewed from any device with Wi-Fi capabilities

Fully Integrated System Design:

- Integrated PCB (w/ ESP32) for motor controls, power, distance sensors, and navigation
- Microphone PCB on top layer (not displayed)

Project Timeline



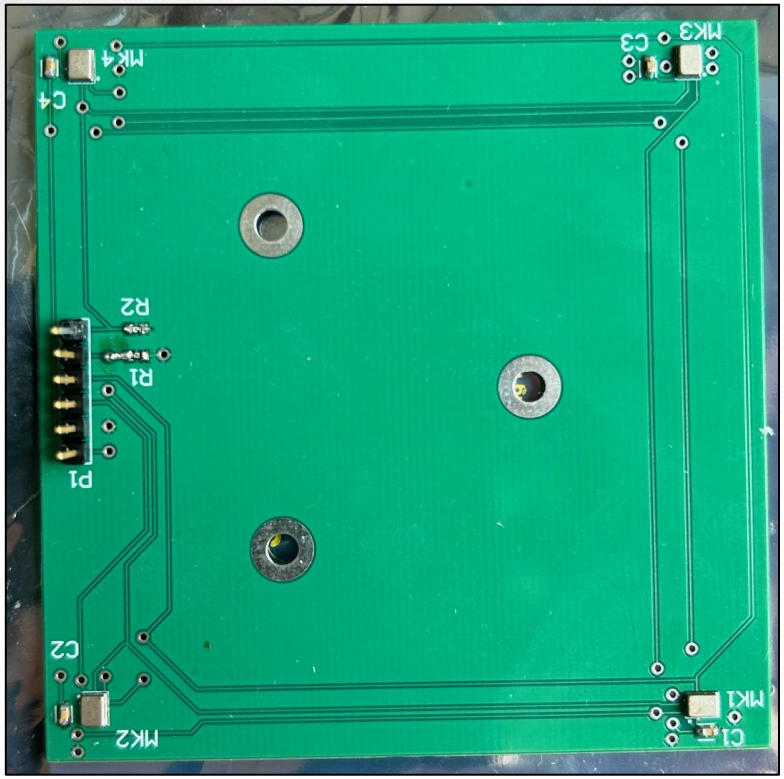
Input Signal Software

Samuel Lackey

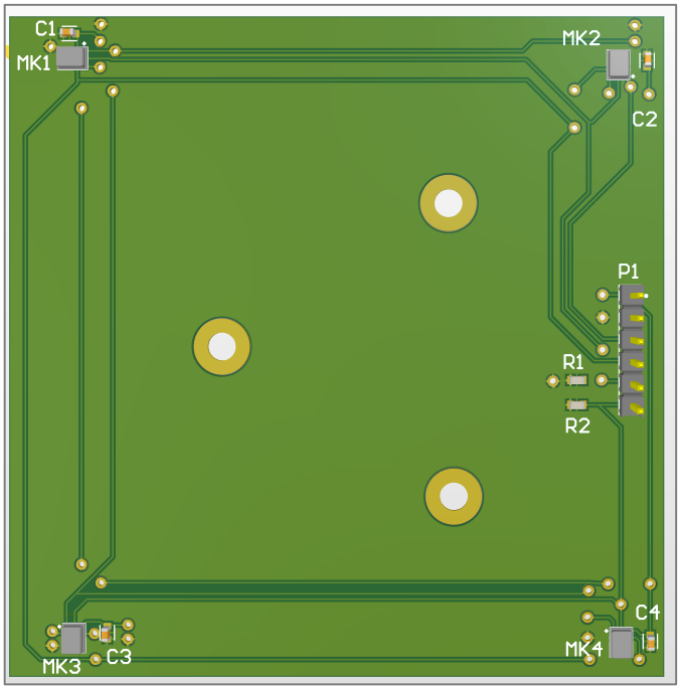
Accomplishments since last update : 35 Hours of Effort	Ongoing progress/problems and plans until the next presentation
• Soldered on Microphone PCB • Tested Algorithm on new PCB • Integrated algorithm into team's code	• Continue testing, and improving • Not very accurate with current implementation – Need to smooth more for direction

Input Signal Software

Samuel Lackey



Microphone Array PCB



Microphone Array PCB 3D Design

All Mics are now working as expected

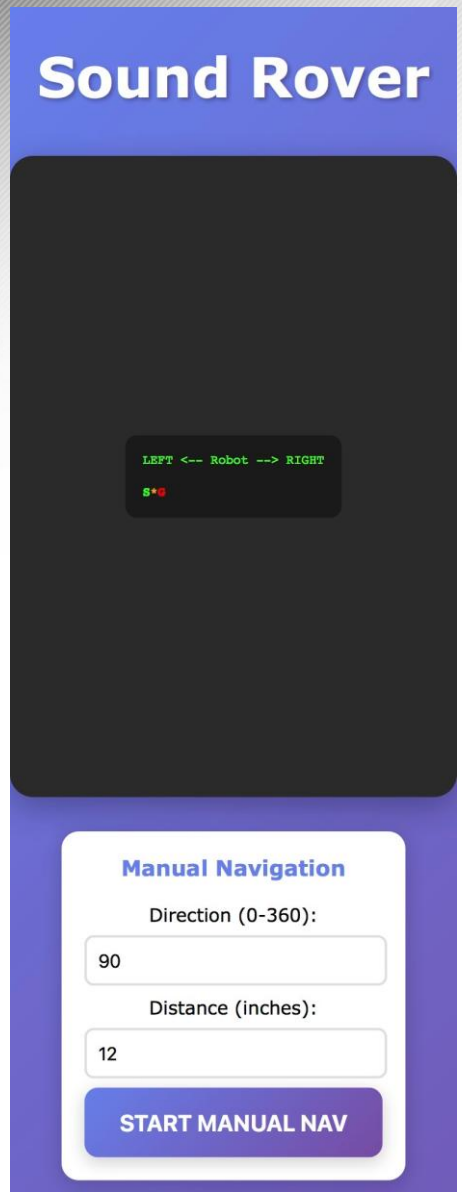
Mechanical/ Motor Controls Subsystem

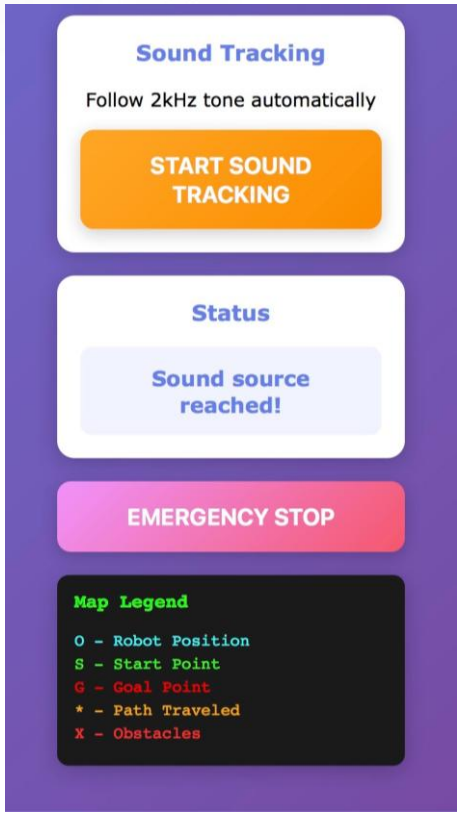
Sanjana Pasumarthi

Accomplishments since Last Update 25 hours of effort	Ongoing progress/problems and plans until the next presentation
• Sub-system is complete, all systems are integrated • Reprinting of 3D parts for better fit of the parts • Testing optimal pauses and timing of movement • More progress on UI, still making changes to certain visuals	• Improve the accuracy of the robot and bring it to its final settings • Program the robot to be able to return to its original location

Mechanical/ Motor Controls Subsystem

Sanjana Pasumarthi





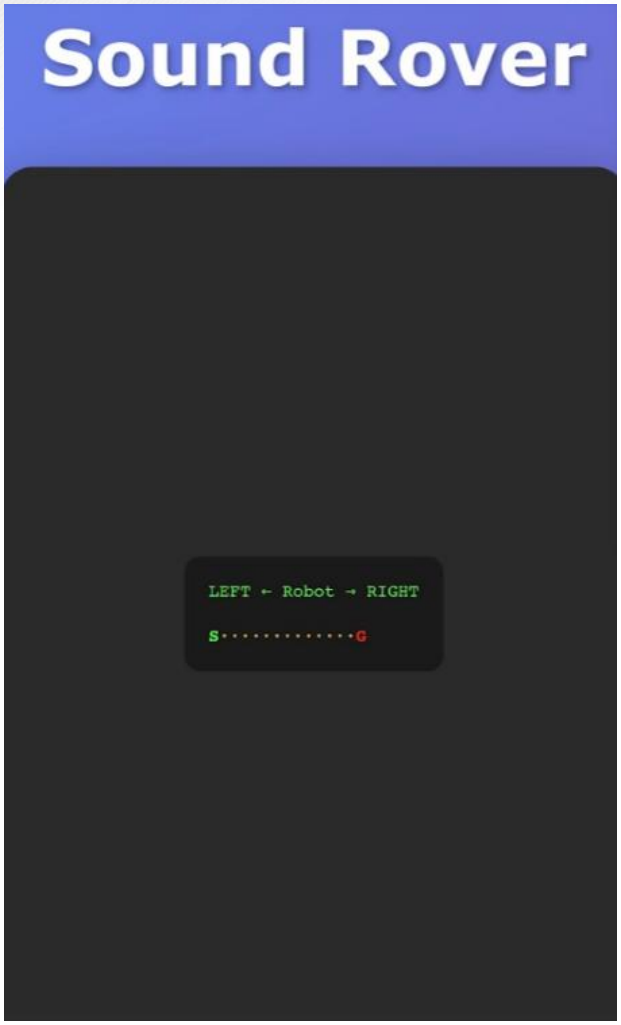
- Final UI design of website accessible through connect of ESP Wi-Fi module
- Constantly updated map after every node
- Some things will be removed in the final design, as they are used for just the testing process
- Phone View

Navigation Software Subsystem

Lucas Kettler

Accomplishments since Last Update 25 hrs of effort	Ongoing progress/problems and plans until the next presentation
• Completed/Validated integrations	• Improve accuracy of integrated system • Perform debugging

Navigation Software Subsystem
Lucas Kettler



- 90 degrees, 126 inches away destination input (Go right 14 nodes)
- Before input software integration
- Space for map expansion

Power & Distance Sensor Subsystem

Joshua Teele

Accomplishments since Last Update 25 hrs of effort	Ongoing progress/problems and plans until the next presentation
• Tested all subsystems on MCU • Attached all subsystems to MCU for integration	• Help other work on fully integrated robot • Fix up wiring and buck converter placement on the chassis to make it look neater

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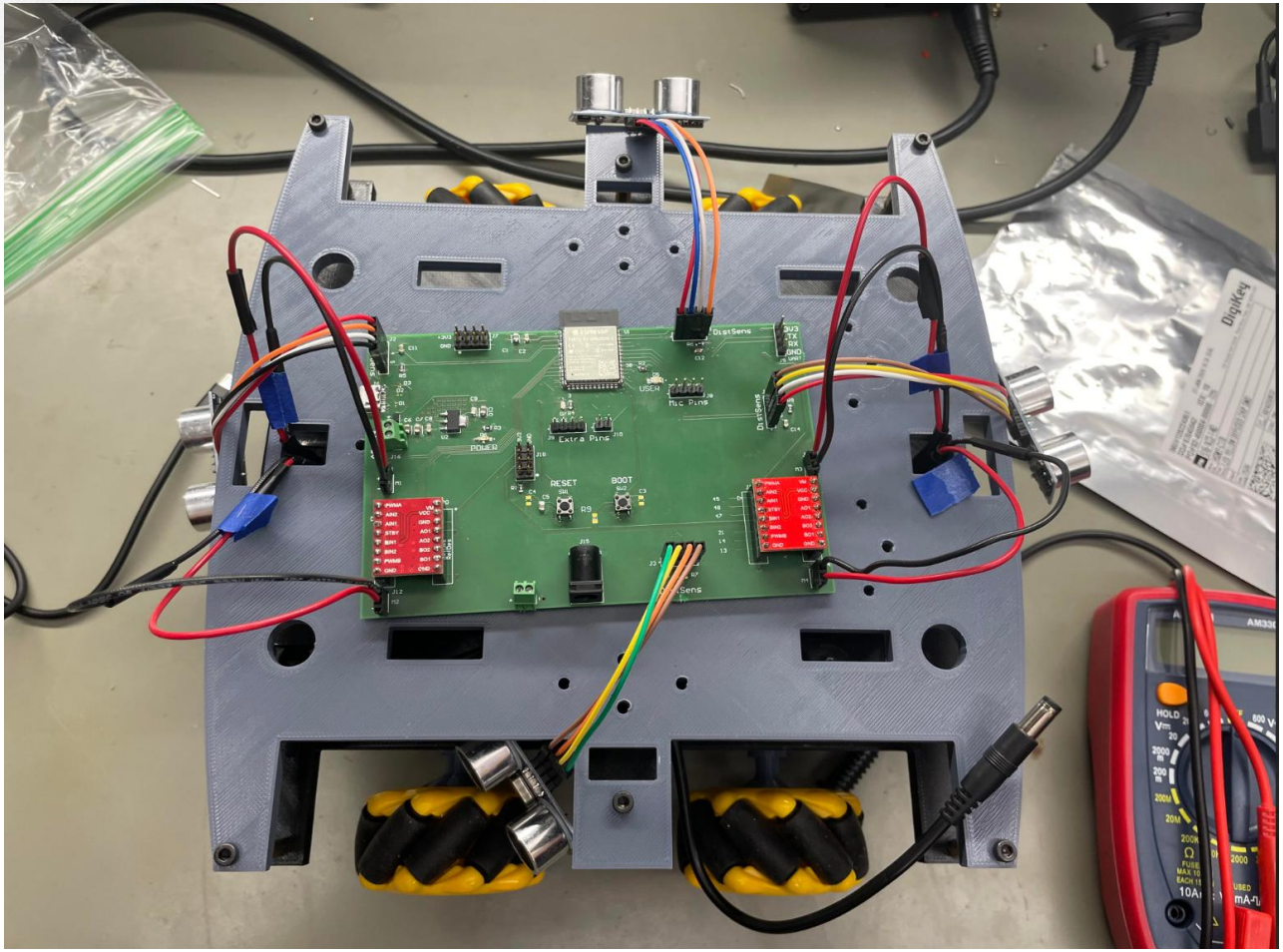
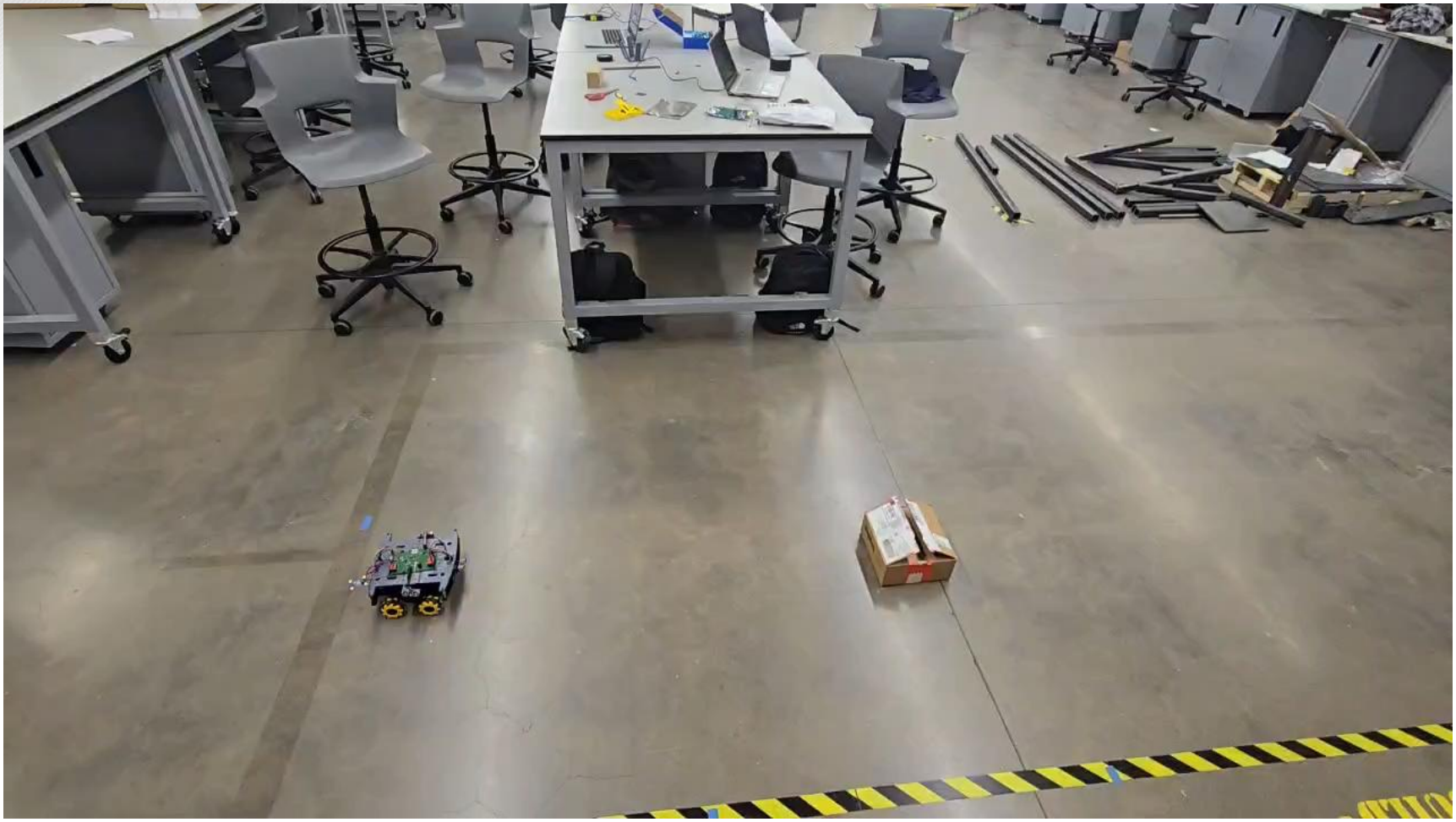
Power & Distance Sensor Subsystem

Joshua Teele



Video of Navigation



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Execution Plan

Tests	1/15/26	1/22/26	1/29/26	2/5/26	2/12/26	2/19/26	2/26/26	3/5/26	3/12/26	3/19/26	3/26/26 - DONE
Motor Controls Subsystem ~ Sanjana Pasumarthi											
Design and 3D Print Proper Wheel Shaft											
Design integrated PCB w/ Josh for power and motor controls											
Design and 3D Print Top Layer											
Integrate Sound Rover Firmware with Navigation & Sound Software & Test Rover											
Navigation Software Subsystem ~ Lucas Kettler											
Integrate with Input Signal Code											
Integrate with Sensor Code											
Integrate with Motor Control Code											
Adapt Grid map for use with web app											
Adapt code from simulation to Real test code											
Power & Distance Sensor Subsystem ~ Joshua Teele											
Design MCU Board											
Solder PCBs											
Integrate Sensor Code with Lucas											
Input Signal Subsystem ~ Samuel Lackey											
Design Microphone Array PCB											
Integrate Input Signal Software with Navigation											
Finalize all Software Integration											
Solder and Connect PCB to Team system											

Validation Plan

Paragraph	Test Name	Success Criteria	Methodology	Status	Responsible Engineer(s)
3.2.1.1	DOA Angle Accuracy	All 5 tests must log a detected angle within an 3.6°	Place speaker playing 450Hz tone at a known angle, and 0.5m distance. Run the detection test 5 times	UNTESTED	Sam
3.2.1.2	DOA With Background Noise	All 5 tests must log a detected angle within an 3.6°	Place speaker playing 450Hz tone at a known angle, and 0.5m distance while separate 550Hz tone is playing. Repeat 5 times	UNTESTED	Sam
3.2.1.3	Obstacle Detection	Verify robot halts movement at the node prior to an object	Isolated test with robot attempting to drive to a goal behind an obstacle	TESTED	Lucas, Joshua, Sanjana
3.2.1.4	Grid Mapping	Verify the robot updates a grid map for the web app	View output of webapp upon completion of path	TESTED	Lucas, Sanjana
3.2.1.5	Webapp	The webapp must show the angle, estimated distance, and control the robot.	Run the program and connect to wifi, trigger a sound, access the webapp, and move.	TESTED	Sam, Sanjana
3.2.1.6	Distance Estimation	The estimated distance must be less than 0.5m of the true distance	Run the distance calibration software, move sound source from 0.5 to 1m to 2m.	UNTESTED	Sam
3.2.2.1	Accurate Robot Movement	Robot moves in intended direction determined by program	Series of coordination tests for the robot to move to the sound source	TESTED	Full Team
3.2.2.2	Constant Response rate of robot	Robot has a constant rate of response with +/- 5% of average time	Multiple trials of measuring time it takes for rover to change status	UNTESTED	Full Team
3.2.2.3	Structurally strong Robot	Wheel shaft does not break easily, no unintended movement from motors	Test robot with obstacles and quick movement	UNTESTED	Sanjana
3.2.3.3.1	Input Voltage(USB)	Ensure voltage at USB port is above 4.75V (minimum needed voltage for linear regulator)	Use multimeter to test output of buck converter	TESTED	Joshua
3.2.3.3.2	Input Voltage(DC jack)	Ensure voltage at DC jack is between 10.8 and 13.2 V (+/-10% of motor voltage)	Use multimeter to test output of DC jack	TESTED	Joshua
N/A	Full System Demo	Robot is able to navigate autonomously around obstacles, reach goal, and display data on app	Multiple observations and trials of robot	TESTED	Full Team

Thank You,
Any Questions?

Sound Localization Robot 5 – Team 23

Lucas Kettler, Samuel Lackey, Joshua Teele,
Sanjana Pasumarthi